Physics (1)

Code :EBA1110

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Grading Strategies

5th exam in fifth week	10 pts.
Assignments & Quizzes	5 pts.
7th exam in seventh week	15 pts.
12th exam in twelfth week	10 pts.
Assignments & Quizzes	10 pts.
Lab	10 pts.
Final exam	40 pts.
Total points	100 pts.

Course Content

- Lecture 1 : Introduction to Vectors.
- Lecture 2 : Newton Laws of motion.
- Lecture 3 :Work and Energy.
- Lecture 4 : Introduction to Waves.
- Lecture 5: Sound Waves, Velocity of sound in solid, liquid and gases.
- Lecture 6 : Superposition of waves.
- Lecture 7 : Capacitors.
- Lecture 8 : Electric Current and Resistance.
- Lecture 9 : Kirchhoff Law and RC circuit.
- Lecture 10 : Magnetic force , Magnetic field.
- Lecture 11 : Source of Magnetic field.
- Lecture 12 : Faraday Law, Inductance.

Textbook : Physics for Scientist and Engineers with Modern physics

LECTURE 1

Introduction to Vectors

Vectors and scalars:

Scalar Quantity

Many quantities in geometry and physics represented by a single real number.

Example : time, temperature, mass , money.

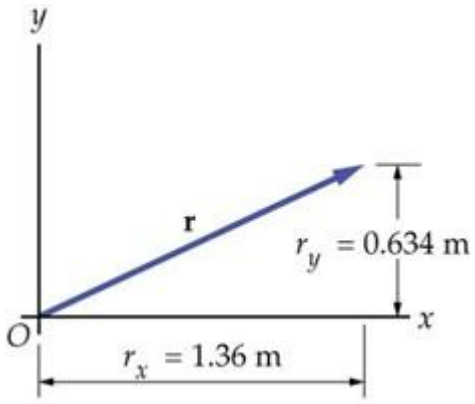
Vector Quantity

Other quantities cannot be completely characterized by a single real number but involve both magnitude and direction.

Example : force , velocity, acceleration.

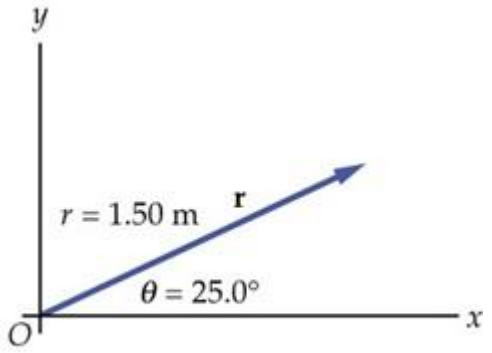
Vector Representation

In two dimensions used Cartesian Form



$(r_x, r_y) = (1.36, 0.634)$

In plane used Polar Form



$\vec{A} = |A| \langle \theta^o$

The vector A in general Cartesian form

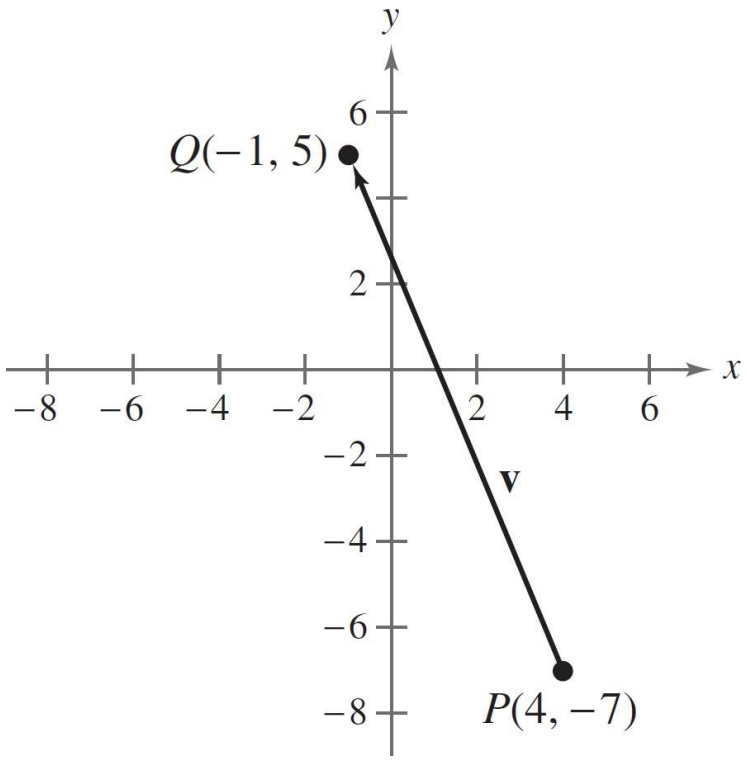
$$\vec{A} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$$

The magnitude vector A

$$|A| = \sqrt{x^2 + y^2 + z^2}$$

Example :

Find the component form and magnitude of the vector **v** that has initial point (4, –7) and terminal point (–1,5).



Let P(4,-7)

and Q(-1,5)

as shown in Figure

Then $\vec{v} = \overrightarrow{PQ} = Q - P$

$\vec{v} = (-1-4\ ,\ 5+7)$ $\vec{v} = (-5,12)$

and the magnitude is $|v| = \sqrt{(-5)^2 + (12)^2}$ $\sqrt{} = 169$ 13

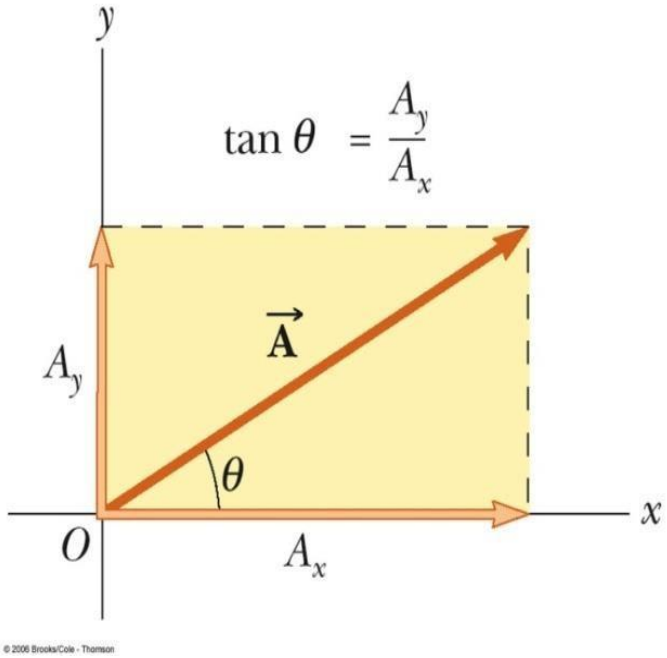
From Polar to Cartesian coordinates and vice versa:

- The x-component of a vector is the projection along the x-axis

$$Cos\,\theta = \frac{A_x}{A} \qquad A_x = A\,\cos\theta$$

- The y-component of a vector is the projection along the y-axis

$$Sin\,\theta = \frac{A_y}{A} \qquad A_y = A\,\sin\theta$$



- Then, the direction $\tan\theta = \frac{A_y}{A_x}$

$$\theta = \tan^{-1}\left(\frac{A_y}{A_x}\right)$$

- And the magnitude $|\bar{A}| = \sqrt{{A_x}^2 + {A_y}^2}$

Example

Find the Cartesian form of $\vec{A} = 55 \langle 205^o$

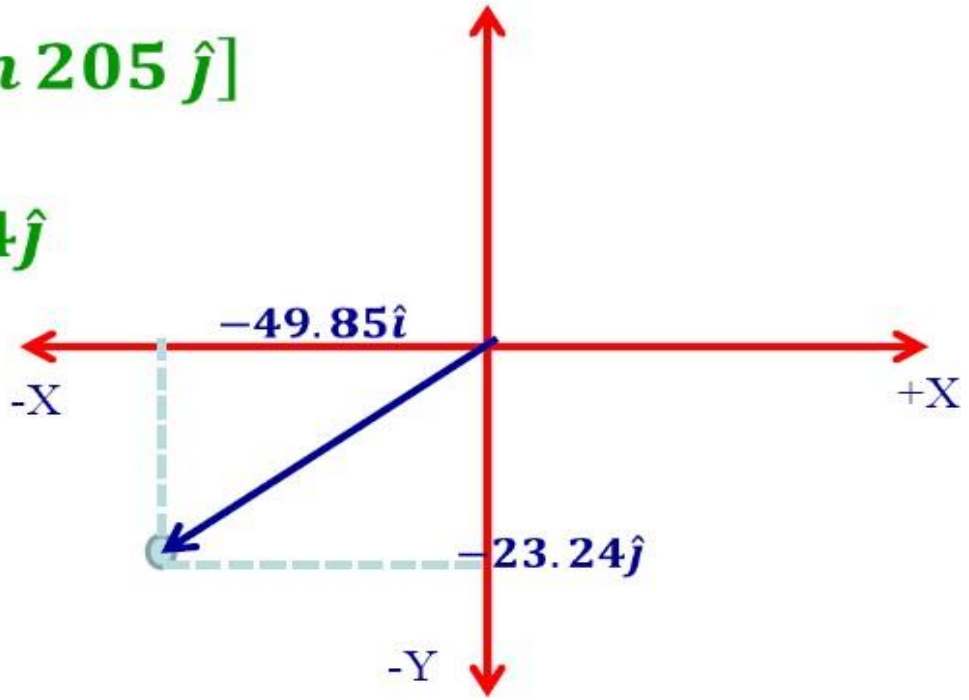
Solution

In general :

$$\vec{A} = |A| \, [cos\,\theta \, \hat{\imath} + sin\,\theta \, \hat{\jmath}]$$

$$\vec{A} = 55 \, [cos\,205 \, \hat{\imath} + sin\,205 \, \hat{\jmath}]$$

$$\vec{A} = -49.85\hat{\imath} \; - 23.24\hat{\jmath}$$



Example

Find the polar form of $\vec{A} = -3\hat{\imath} + 4\hat{\jmath}$,

Solution

$$A = \sqrt{x^2 + y^2} = \sqrt{(-3)^2 + (4)^2} = 5$$

To find the angle by ⟹ $\tan\theta = \frac{y}{x}$

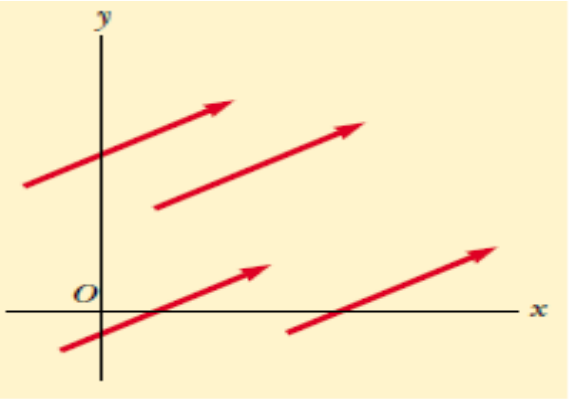
$$\tan\theta = \frac{4}{-3} \Longrightarrow \theta = \begin{cases} -53.13^o & \times \\ 126.87^o & \checkmark \end{cases}$$

The vector in polar form ⟹ $\vec{A} = 5\langle 126.87^o$

Some properties of vectors:

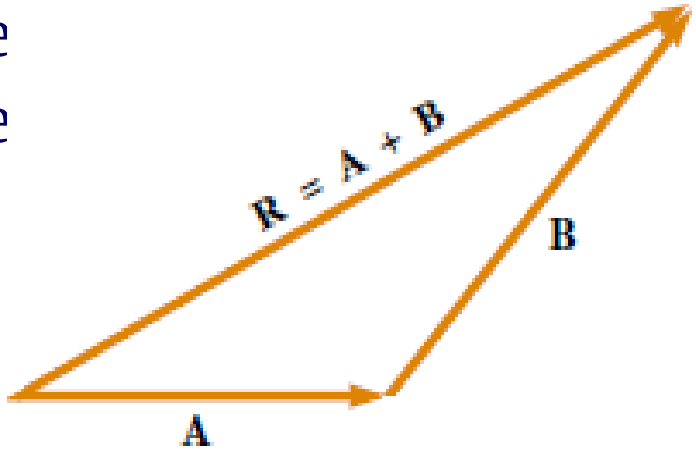
1- Equality of two vectors

- $\vec{A}$ and $\vec{B}$ will be equal if they equal in magnitude and they were in the same direction.



2- Addition in vectors

To add vector $\vec{B}$ to vector $\vec{A}$, first draw vector $\vec{A}$, and then draw vector $\vec{B}$ to the same scale with its tail starting from the tip of $\vec{A}$, as shown in Figure. The resultant vector $\vec{R}= \vec{A} + \vec{B}$ is the vector drawn from the tail of $\vec{A}$ to the tip of $\vec{B}$



3- Negativity of vectors

The negative of the vector $\overrightarrow{A}$ is defined as the vector that when added to $\overrightarrow{A}$ gives zero for the vector sum. That is, $\overrightarrow{A} + (-\overrightarrow{A}) = 0$.

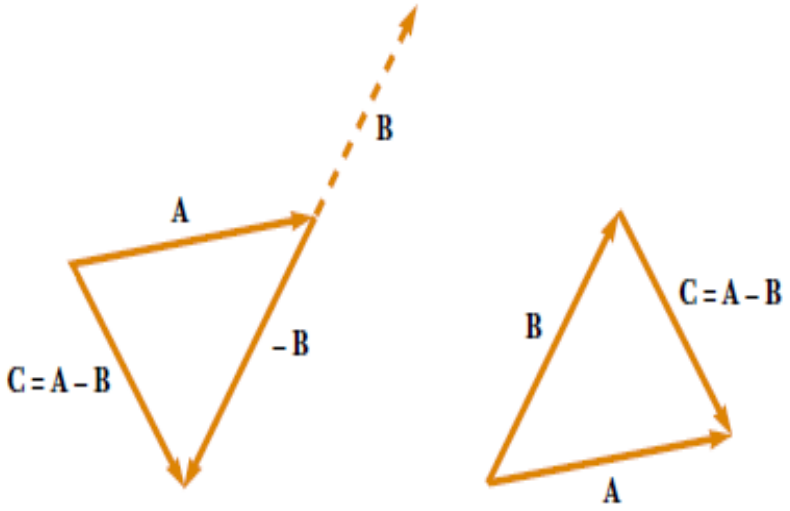
The vector $\overrightarrow{A}$ and $-\overrightarrow{A}$ have the same magnitude but point in opposite directions.

4- Subtraction of vectors

The operation of vector subtraction makes use of the definition of the negative of a vector.

$\overrightarrow{A} - \vec{B}$ as vector $-\vec{B}$ added to vector $\overrightarrow{A}$.

$\overrightarrow{A} - \vec{B} = \overrightarrow{A} + (\overrightarrow{-B})$



5- Vector Multiplication

* Cross Product

The cross product **a** × **b** (read "a cross b"), is a vector that is perpendicular to both **a** and **b**, and thus normal to the plane containing them.

definition

$$\vec{a} \times \vec{b} = |\vec{a}|\,|\vec{b}| \sin\theta\, \hat{n}$$

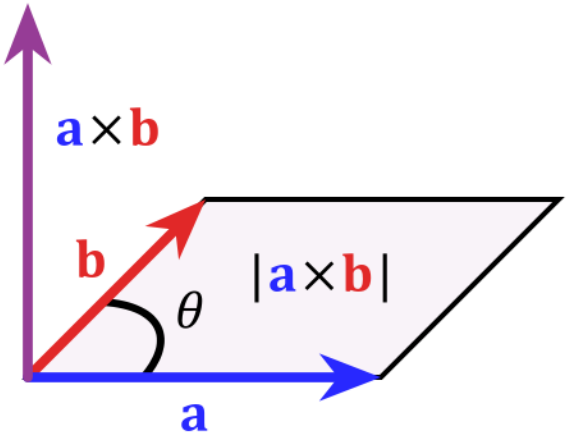
$$\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k} \qquad \vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$$

$$\vec{a} \times \vec{b} = i\,(a_2b_3 - a_3b_2) - j\,(a_1b_3 - a_3b_1) + k\,(a_1b_2 - a_2b_1)$$

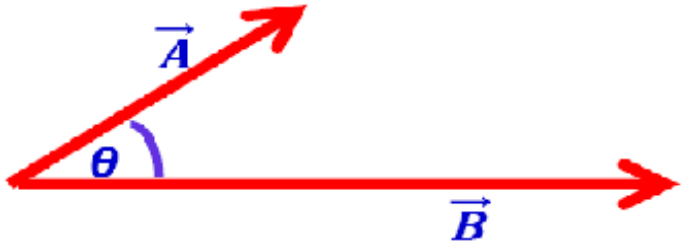
$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$i \times j = k$ $\quad j \times k = i$ $\quad k \times i = j$

$j \times i = -k$ $\quad k \times j = -i$ $\quad i \times k = -j$



** Dot Product



definition $\vec{A}.\vec{B} = |A||B| \cos\theta$

If $\vec{A} = x_1\hat{\imath} + y_1\hat{\jmath} + z_1\hat{k}$

$\vec{B} = x_2\hat{\imath} + y_2\hat{\jmath} + z_2\hat{k}$

Rule $\vec{A}.\vec{B} = x_1x_2 + y_1y_2 + z_1z_2$

As we know $\hat{\imath}.\hat{\imath} = \hat{\jmath}.\hat{\jmath} = \hat{k}.\hat{k} = 1$

and $\hat{\imath}.\hat{\jmath} = \hat{\jmath}.\hat{k} = \hat{k}.\hat{\imath} = 0$

Example

Find the angle between $\vec{A} = -10\hat{\imath} - 25\hat{\jmath} - 50\hat{k}$ *and* $\vec{B} = -12\hat{\imath} + 20\hat{\jmath}$

Solution

rule $\quad \vec{A}.\vec{B} = |A||B| \cos\theta$

since $\quad \cos\theta = \dfrac{\vec{A}.\vec{B}}{|A||B|}$

then $\quad \cos\theta = \dfrac{(-10\hat{\imath} - 25\hat{\jmath} - 50\hat{k}).(-12\hat{\imath} + 20\hat{\jmath})}{\sqrt{100 + 625 + 2500}\sqrt{144 + 400}}$

then $\quad \cos\theta = \dfrac{(120 - 500)}{(56.8)(23.3)} = \dfrac{-380}{1324.8} = -0.2868$

then $\quad \theta = 106.67^{o}$

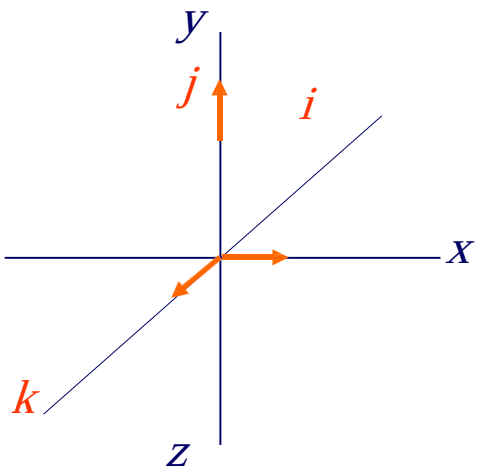
Unit Vectors

- A unit vector is simply a vector of unit length with the same direction as a given vector.

Rule $\hat{n} = \dfrac{\vec{A}}{|\vec{A}|}$

- which has unit magnitude and is in the direction of $\vec{A}$.
- The unit vectors in the positive direction of x, y and z axis are labeled as $\hat{i}$, $\hat{j}$ and $\hat{k}$ respectively with a small hat on top.
- Then we can express the vector $\vec{A}$ as,

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$



Example

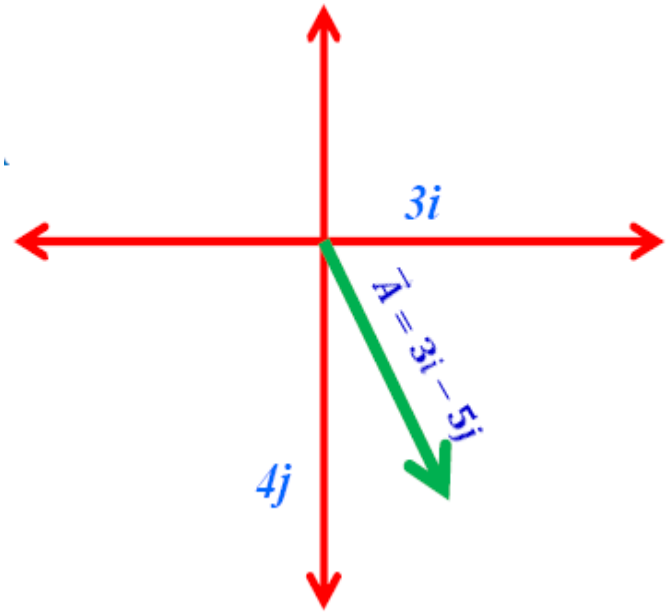
- Find the unit vector of the following Cartesian vector A

$$\vec{A} = 3i - 5j$$

Solution

$$\hat{A} = \frac{\vec{A}}{|A|}$$

$$\hat{A} = \frac{3i - 5j}{\sqrt{9 + 25}} = \frac{3i - 5j}{5.83} = 0.515i - 0.858j$$




Thank
You